Experiment 26: Traffic Squad – Penalty Collection & Management Web Application

Aim:

To develop an interactive web application that enables traffic police to create online FIRs and traffic challans with vehicle-owner details, proof photographs, and fine information. Users can view their challans and filed cases, while administrators manage police logins and daily reports.

Algorithm:

1. Create a responsive website with separate sections for User, Police, and Admin.
2. Allow users to register, log in, and view their traffic challans and filed cases.
3. Provide a police login interface to access traffic rules and manage violations.
4. Allow police to create a case by entering vehicle-owner details, violation information, fine amount, and proof photo.
5. Generate a challan receipt containing the case number and fine details.
6. Display recent traffic reports and provide a link to Google Maps.
7. Allow the administrator to manage police logins and view daily reports.
8. Store and display demonstration records using browser local storage.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Traffic Squad | Challan Management</title>

<style>

:root{--bg:#0b1220;--panel:#141f32;--line:#2a3a52;--text:#edf4ff;--muted:#9baec8;--blue:#60a5fa;--green:#34d399;--red:#fb7185}

*{box-sizing:border-box}
```

```
body{margin:0;font-family:Arial,sans-serif;background:var(--bg);color:var(--text)}

header{padding:22px 5%;background:#101b2d;border-bottom:1px solid var(--line);display:flex;justify-content:space-between;align-items:center;gap:15px;flex-wrap:wrap}

.brand{font-size:23px;font-weight:bold}.brand span{color:var(--blue)}

header small{color:var(--muted)}

nav{display:flex;gap:8px;flex-wrap:wrap;padding:14px 5%;background:#0e1929;position:sticky;top:0;z-index:2}

button,.btn{border:0;border-radius:9px;padding:11px 15px;cursor:pointer;font-weight:bold;background:var(--blue);color:#071426}

nav button{background:var(--panel);color:var(--text);border:1px solid var(--line)}

nav button:hover,button:hover{filter:brightness(1.15)}

main{max-width:1150px;margin:auto;padding:24px 5%}

.hero{padding:28px;border:1px solid var(--line);border-radius:18px;background:linear-gradient(120deg,#172b48,#111c2f)}

.hero h1{font-size:clamp(26px,5vw,42px);margin:8px 0}

.hero p,.muted{color:var(--muted);line-height:1.6}

.tag{display:inline-block;background:#1d3859;color:#93c5fd;padding:6px 10px;border-radius:20px;font-size:12px}

.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(230px,1fr));gap:15px;margin:20px 0}

.card{background:var(--panel);border:1px solid var(--line);border-radius:15px;padding:19px;min-width:0}

.card h3{margin-top:0}

.metric{font-size:30px;font-weight:bold;margin:10px 0}

label{display:block;font-size:13px;color:#c3d2e7;margin:12px 0 6px}

input,select,textarea{width:100%;padding:12px;background:#0b1627;color:var(--text);border:1px solid #344760;border-radius:8px}

textarea{min-height:75px;resize:vertical}

form button{margin-top:15px}

.row{display:grid;grid-template-columns:1fr 1fr;gap:12px}
```

```
.hidden{display:none!important}

table{width:100%;border-collapse:collapse;margin-top:12px;font-size:13px}

th,td{text-align:left;padding:11px;border-bottom:1px solid var(--line);vertical-align:top}

th{color:#93c5fd}

.table-wrap{overflow-x:auto}

.status{padding:5px 8px;border-radius:12px;background:#163a31;color:#6ee7b7;font-size:11px}

.receipt{border:1px dashed var(--blue);padding:20px;border-radius:12px;margin-top:15px;background:#0c192b}

.receipt h3{color:var(--green)}

#photoPreview{max-width:100%;max-height:220px;border-radius:10px;margin-top:10px}

footer{text-align:center;color:var(--muted);padding:25px;border-top:1px solid var(--line)}

.notice{padding:12px;border-radius:9px;background:#342b18;color:#fcd98a;font-size:13px;line-height:1.5}

.danger{background:#7f1d1d;color:white}

.secondary{background:#263750;color:white}

@media(max-width:600px){.row{grid-template-columns:1fr}header{align-items:flex-start}main{padding-top:15px}th,td{padding:8px}}

</style>

</head>

<body>

<header>

 <div class="brand">🚦 TRAFFIC <span>SQUAD</span><br>

  <small>Penalty Collection &amp;amp; Case Management</small>

 </div>

 <small id="clock">Loading time...</small>

</header>
```

```
<nav>
  <button onclick="showPage('home')">Dashboard</button>
  <button onclick="showPage('user')">User / Challan</button>
  <button onclick="showPage('police')">Police Portal</button>
  <button onclick="showPage('admin')">Admin Reports</button>
  <button onclick="showPage('map')">Traffic Map</button>
</nav>

<main>
<section id="home">
  <div class="hero">
    <span class="tag">DIGITAL TRAFFIC MANAGEMENT</span>
    <h1>Safer roads. Smarter enforcement.</h1>
    <p>Create demo traffic cases, attach proof photographs, issue challan receipts and view reports from one dashboard.</p>
    <button onclick="showPage('police')">Create Traffic Case →</button>
    <button class="secondary" onclick="showPage('user')">Find My Challan</button>
  </div>
  <div class="grid">
    <div class="card"><span class="muted">Total demo cases</span><div class="metric" id="total">0</div></div>
    <div class="card"><span class="muted">Total fine amount</span><div class="metric" id="fine">₹0</div></div>
    <div class="card"><span class="muted">Cases today</span><div class="metric" id="today">0</div></div>
  </div>
  <div class="card">
    <h3>Recent Traffic Cases</h3>
    <div id="recent" class="table-wrap"></div>
```

```
 </div>
</section>

<section id="user" class="hidden">
 <h2>👤 User Portal</h2>
 <div class="grid">
  <div class="card">
   <h3>Find Challan / Filed Case</h3>
   <p class="muted">Enter the vehicle number or challan ID used in the demonstration.</p>
   <form onsubmit="findCases(event)">
    <label>Vehicle Number or Challan ID</label>
    <input id="lookup" placeholder="TN01AB1234 or TS-1001" required>
    <button type="submit">Search Records</button>
   </form>
  </div>
  <div class="card">
   <h3>My Challan Results</h3>
   <div id="userResults" class="muted">Your search results will appear here.</div>
  </div>
 </div>
</section>

<section id="police" class="hidden">
 <h2>👮 Police Portal</h2>
 <div class="notice">Student demo: this portal does not authenticate real police officers or create legally valid FIRs.</div>
 <div class="grid">
```

```
  <div class="card">
   <h3>Create Traffic Case</h3>
   <form id="caseForm" onsubmit="createCase(event)">
    <label>Vehicle Owner Name</label>
    <input id="owner" required placeholder="Owner name">
    <div class="row">
     <div><label>Vehicle Number</label><input id="vehicle" required placeholder="TN01AB1234"></div>
     <div><label>Violation</label>
      <select id="violation">
       <option>Helmet violation</option>
       <option>Signal jumping</option>
       <option>Speeding</option>
       <option>No parking</option>
       <option>Seat belt violation</option>
       <option>Other traffic violation</option>
      </select>
     </div>
    </div>
    <div class="row">
     <div><label>Fine Amount (₹)</label><input id="amount" type="number" min="0" required value="500"></div>
     <div><label>Case Date</label><input id="caseDate" type="date" required></div>
    </div>
    <label>Location / Road</label>
    <input id="location" required placeholder="Road or junction">
    <label>Case Description</label>
    <textarea id="description" placeholder="Enter violation details"></textarea>
    <label>Proof Photograph (optional)</label>
```

```
      <input id="photo" type="file" accept="image/*" onchange="previewPhoto(event)">
      <img id="photoPreview" class="hidden" alt="Proof photo preview">
      <button type="submit">Generate Demo Challan</button>
    </form>
    <div id="caseMessage"></div>
  </div>
  <div class="card">
    <h3>Case Management</h3>
    <p class="muted">Review and remove demonstration records.</p>
    <div id="policeCases" class="table-wrap"></div>
  </div>
 </div>
</section>

<section id="admin" class="hidden">
 <h2> 🛡️ Admin Dashboard</h2>
 <div class="notice">For demonstration only. A production system requires secure administrator authentication and access controls.</div>
 <div class="card">
  <h3>Daily Traffic Report</h3>
  <button onclick="exportCSV()">Export CSV Report</button>
  <div id="adminCases" class="table-wrap"></div>
 </div>
 <div class="card" style="margin-top:16px">
  <h3>Police Login Management — Demo</h3>
  <p class="muted">This prototype displays a sample police account record. It does not manage real accounts.</p>
  <table><tr><th>Officer ID</th><th>Role</th><th>Status</th></tr>
```

```
  <tr><td>DEMO-POLICE-01</td><td>Traffic Officer</td><td><span class="status">Sample account</span></td></tr></table>
 </div>
</section>

<section id="map" class="hidden">
 <h2>🗺️ Traffic Location Map</h2>
 <div class="card">
  <p>Open a map to inspect the location entered in a case. This link opens a general map search, not a live traffic feed.</p>
  <form onsubmit="openMap(event)">
   <label>Road, junction or location</label>
   <input id="mapLocation" placeholder="e.g. Anna Salai, Chennai" required>
   <button type="submit">Open Map</button>
  </form>
  <p class="muted">Live traffic data depends on the map provider and is not collected by this prototype.</p>
 </div>
</section>
</main>

<footer>Traffic Squad · Internet Programming Lab · Educational prototype</footer>

<script>
const KEY = "trafficSquadDemoCases";
let proofImage = "";

function getCases(){
 try{return JSON.parse(localStorage.getItem(KEY) || "[]")}
```

```
 catch{return []}
}
function saveCases(data){
 localStorage.setItem(KEY, JSON.stringify(data));
}
function escapeHTML(value){
 return String(value ?? "").replace(/[&<>"']/g, ch => ({
  "&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;",'"':"&#39;"
 }[ch]));
}
function showPage(id){
 document.querySelectorAll("main section").forEach(s=>s.classList.add("hidden"));
 document.getElementById(id).classList.remove("hidden");
 refresh();
}
function refresh(){
 const cases=getCases();
 document.getElementById("total").textContent=cases.length;

document.getElementById("fine").textContent="₹"+cases.reduce((sum,c)=>sum+Number(c.amount||0),0).toLocaleString("en-IN");
 const today=new Date().toISOString().slice(0,10);

document.getElementById("today").textContent=cases.filter(c=>c.date===today).length;


 const table=(rows,admin=false)=>rows.length?
  `<table><tr><th>Challan</th><th>Vehicle / Violation</th><th>Fine</th>${admin?"<th>Date</th>":""}<th>Action</th></tr>`+
```

```
  rows.map(c=>`<tr>
    <td>${escapeHTML(c.id)}</td>
    <td>${escapeHTML(c.vehicle)}<br>${escapeHTML(c.violation)}</td>
    <td>₹${Number(c.amount).toLocaleString("en-IN")}</td>
    ${admin?`<td>${escapeHTML(c.date)}</td>`:""}
    <td><button class="secondary" onclick="receipt('${c.id}')">Receipt</button>
    <button class="danger" onclick="removeCase('${c.id}')">Delete</button></td>
  </tr>`).join("")+"</table>":"<p class='muted'>No cases available.</p>";

document.getElementById("recent").innerHTML=table(cases.slice().reverse().slice(0,5));
 document.getElementById("policeCases").innerHTML=table(cases.slice().reverse());

document.getElementById("adminCases").innerHTML=table(cases.slice().reverse(),true);
}
function previewPhoto(event){
 const file=event.target.files[0];
 const img=document.getElementById("photoPreview");
 proofImage="";
 if(!file){img.classList.add("hidden");return}
 if(!file.type.startsWith("image/")){alert("Please choose an image.");event.target.value="";return}
 if(file.size>700*1024){
  alert("For this demo, choose an image smaller than 700 KB.");
  event.target.value="";img.classList.add("hidden");return;
 }
 const reader=new FileReader();
```

```
 reader.onload=()=>{
  proofImage=reader.result;
  img.src=proofImage;img.classList.remove("hidden");
 };
 reader.readAsDataURL(file);
}
function createCase(event){
 event.preventDefault();
 const cases=getCases();
 const id="TS-"+(1001+Date.now()%8999);
 const record={
  id,owner:document.getElementById("owner").value.trim(),
  vehicle:document.getElementById("vehicle").value.trim().toUpperCase(),
  violation:document.getElementById("violation").value,
  amount:Number(document.getElementById("amount").value),
  date:document.getElementById("caseDate").value,
  location:document.getElementById("location").value.trim(),
  description:document.getElementById("description").value.trim(),
  photo:proofImage
 };
 cases.push(record);saveCases(cases);
 document.getElementById("caseMessage").innerHTML=
  `<div class="receipt"><h3>✓ Demo Challan Created</h3>
  <p>Challan ID: <b>${escapeHTML(id)}</b></p>
  <p>Vehicle: ${escapeHTML(record.vehicle)}</p>
  <p>Fine: ₹${record.amount.toLocaleString("en-IN")}</p>
  <button onclick="receipt('${id}')">View Receipt</button></div>`;
 document.getElementById("caseForm").reset();
```

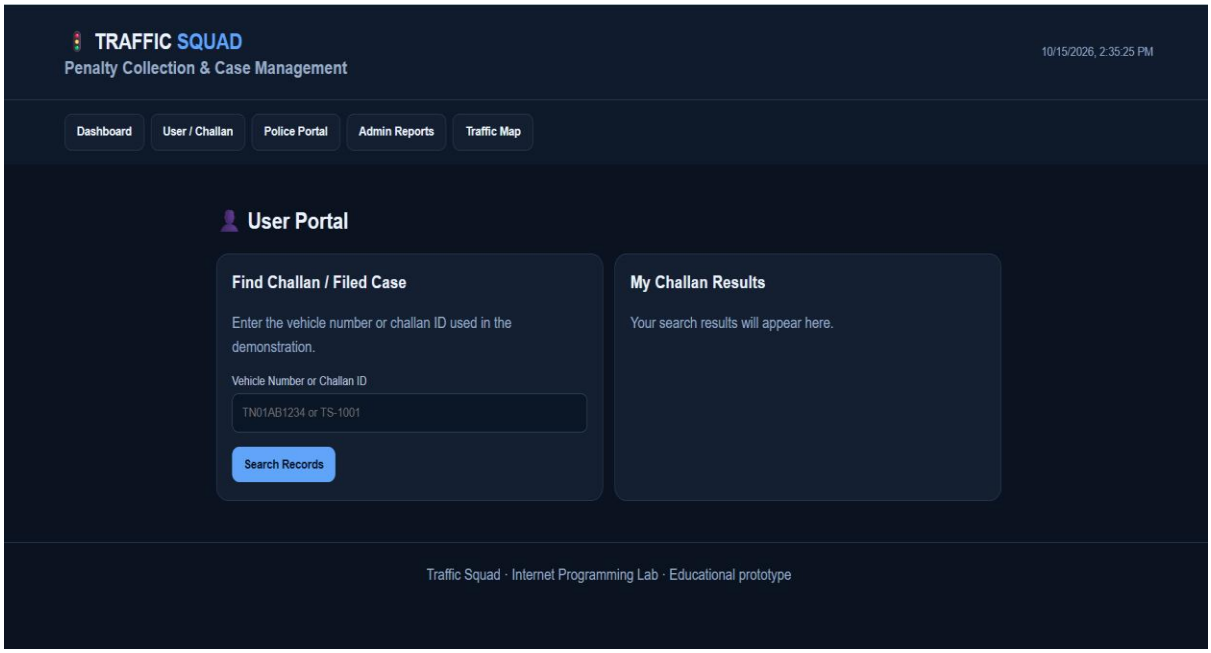
```
 document.getElementById("caseDate").value=new Date().toISOString().slice(0,10);
 document.getElementById("photoPreview").classList.add("hidden");proofImage="";
 refresh();
}
function findCases(event){
 event.preventDefault();
 const query=document.getElementById("lookup").value.trim().toUpperCase();
 const matches=getCases().filter(c=>c.vehicle.toUpperCase()===query||c.id.toUpperCase()===query);
 const out=document.getElementById("userResults");
 if(!matches.length){out.textContent="No matching demonstration record found.";return}
 out.innerHTML=matches.map(c=>`<div class="receipt">
  <h3>Traffic Challan</h3><p><b>ID:</b> ${escapeHTML(c.id)}</p>
  <p><b>Owner:</b> ${escapeHTML(c.owner)}</p>
  <p><b>Vehicle:</b> ${escapeHTML(c.vehicle)}</p>
  <p><b>Violation:</b> ${escapeHTML(c.violation)}</p>
  <p><b>Fine:</b> ₹${Number(c.amount).toLocaleString("en-IN")}</p>
  <p><b>Date:</b> ${escapeHTML(c.date)}</p>
  <p><b>Location:</b> ${escapeHTML(c.location)}</p>
  <p><b>Status:</b> Demo record — not an official challan</p>
  <button onclick="receipt('${c.id}')">Print Receipt</button>
 </div>`).join("");
}
function receipt(id){
 const c=getCases().find(x=>x.id===id);if(!c)return;
 const w=window.open("","_blank","width=650,height=750");
 if(!w){alert("Allow pop-ups to print the receipt.");return}
```

```
 w.document.write(`<!DOCTYPE html><html><head><title>Demo Challan</title>
 <style>body{font-family:Arial;padding:30px;color:#222}h1{color:#145da0}
 .box{border:2px dashed #555;padding:20px}img{max-width:250px;max-height:200px}
 @media print{button{display:none}}</style></head><body>
 <h1>TRAFFIC SQUAD</h1><h2>Demonstration Challan Receipt</h2><div class="box">
 <p><b>Challan ID:</b> ${escapeHTML(c.id)}</p>
 <p><b>Owner:</b> ${escapeHTML(c.owner)}</p><p><b>Vehicle:</b> ${escapeHTML(c.vehicle)}</p>
 <p><b>Violation:</b> ${escapeHTML(c.violation)}</p><p><b>Fine:</b> ₹${Number(c.amount).toLocaleString("en-IN")}</p>
 <p><b>Date:</b> ${escapeHTML(c.date)}</p><p><b>Location:</b> ${escapeHTML(c.location)}</p>
 <p><b>Description:</b> ${escapeHTML(c.description)}</p>
 ${c.photo?`<p>Proof photo:</p><img src="${c.photo}" alt="Proof photo">`:""}
 <hr><b>Educational prototype only — not an official legal document.</b></div>
 <button onclick="window.print()">Print / Save as PDF</button></body></html>`);
 w.document.close();
}
function removeCase(id){
 if(!confirm("Delete this demonstration case?"))return;
 saveCases(getCases().filter(c=>c.id!==id));refresh();
}
function exportCSV(){
 const rows=getCases();
 if(!rows.length){alert("No records to export.");return}
 const fields=["id","owner","vehicle","violation","amount","date","location","description"];
 const csv=[fields.join(","),...rows.map(c=>fields.map(k=>`"${String(c[k]??"").replace(/"/g,'""')}"`).join(","))].join("\n");
```

```
 const url=URL.createObjectURL(new Blob([csv],{type:"text/csv;charset=utf-8;"}));
 const a=document.createElement("a");a.href=url;a.download="traffic-squad-demo-report.csv";a.click();
 URL.revokeObjectURL(url);
}
function openMap(event){
 event.preventDefault();
 const place=document.getElementById("mapLocation").value.trim();

window.open("https://www.google.com/maps/search/?api=1&query="+encodeURIComponent(place),"_blank","noopener");
}
function updateClock(){
 document.getElementById("clock").textContent=new Date().toLocaleString();
}
document.getElementById("caseDate").value=new Date().toISOString().slice(0,10);
updateClock();setInterval(updateClock,1000);refresh();
</script>
</body>
</html>
```

Output:



Result

Thus, the Traffic Squad – Penalty Collection & Management Web Application was developed using HTML, CSS, and JavaScript. It demonstrates traffic-case creation, proof-photo preview, challan lookup, printable receipts, map links, and daily report export.